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Multimodal Neonatal Pain Classification Using EEG, ECG, and Respiration Signals with Subject-Level Validation

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Abstract—Neonatal pain assessment is a critical clinical challenge, as preterm infants undergoing repeated procedural pain cannot verbally communicate their discomfort, and existing observational tools such as the Premature Infant Pain Profile (PIPP) are inherently subjective and unsuitable for continuous automated monitoring. This paper proposes a multimodal machine learning framework for objective neonatal pain classification using electroencephalography (EEG), electrocardiography (ECG), and respiration signals. A signal preprocessing pipeline was first validated on the PhysioNet Preterm Infant Cardiorespiratory Signals (PICS) dataset comprising 10 infants, where 26,208 ECG windows were successfully extracted and analysed. Multimodal pain classification was subsequently evaluated on the publicly available UCL neonatal pain dataset comprising 112 neonates and 308 recordings, using an XGBoost classifier with Leave-One-Subject-Out (LOSO) cross-validation. Two evaluation protocols were employed: a broad condition-based scheme and a stricter clinically-validated PIPP-confirmed subset. The proposed framework achieved a mean LOSO AUC of 0.849 on the condition-based cohort (53 evaluable subjects) and 0.929 on the PIPP-confirmed cohort (28 subjects), demonstrating competitive performance for automated multimodal neonatal pain assessment.

Index Terms—Neonatal pain assessment, Multimodal physiological signals, Electroencephalography (EEG), XGBoost, Leave-One-Subject-Out Cross-Validation, PIPP score.

I. INTRODUCTION

A. Background and Motivation

Neonates admitted to Neonatal Intensive Care Units (NICUs) are routinely subjected to numerous painful procedures, including heel lances, venipunctures, immunizations, and endotracheal suctioning. Repeated exposure to untreated or under-treated pain during this critical period of neurodevelopment has been linked to adverse long-term neurodevelopmental outcomes and altered pain perception. Because neonates lack the cognitive and verbal capacity to communicate their pain, clinicians must rely on alternative methods to assess pain accurately. Consequently, the development of reliable, objective pain assessment tools is of paramount importance to ensure adequate analgesia and mitigate the long-term consequences of neonatal pain.

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B. Current Clinical Assessment

In clinical practice, neonatal pain is predominantly assessed using observational scoring systems such as the Premature Infant Pain Profile (PIPP) [1], its revised version (PIPP-R) [2], the Neonatal Infant Pain Scale (NIPS) [3], and the Neonatal Facial Coding System (NFCS) [4]. These scales integrate behavioral cues (e.g., facial grimacing, crying) with physiological indicators (e.g., heart rate, oxygen saturation) and gestational age. While widely adopted, these observational scales possess inherent limitations. They rely heavily on the subjective interpretation of behavioral and physiological indicators by trained clinicians, making them susceptible to inter-observer variability. Furthermore, manual scoring is time-consuming, resource-intensive, and impractical for continuous, real-time monitoring at the bedside. These limitations strongly motivate the development of automated, objective, and multimodal computational frameworks capable of continuous neonatal pain assessment.

C. Machine Learning Approaches

To address the limitations of manual scoring, researchers have increasingly turned to machine learning to decode neonatal pain from physiological signals. Early neurophysiological studies established that nociceptive stimuli elicit distinct cortical responses, such as nociceptive-specific event-related potentials (ERPs) in EEG. Building on this, feature-based machine learning methods have been applied to various physiological modalities, including EEG, ECG, and respiration. Recently, several studies have leveraged the publicly available UCL neonatal dataset to develop automated pain classifiers. For instance, Hamwi *et al.* (2025) [5] demonstrated strong classification performance (AUC 0.94, Accuracy 82%) using Logistic Regression restricted to the oldest gestational age group. Similarly, Rou  t *et al.* (2025) [6] achieved an AUC of 0.90 using Gradient Boosting on the full dataset. While these studies have established strong baselines, critical gaps remain. Most existing pipelines rely on single modalities or lack rigorous subject-independent validation. There is a pressing need for comprehensive frameworks that integrate multiple physiological modalities and are evaluated using strict Leave-

One-Subject-Out (LOSO) cross-validation to ensure robust generalization across unseen patients.

D. Contributions

In this work, the proposed framework is developed using two complementary datasets. The PICS dataset is employed to validate the ECG preprocessing and feature extraction methodology, whereas the publicly available UCL neonatal pain dataset is used for multimodal pain classification and subject-level evaluation.

- Development and validation of an ECG preprocessing and heart-rate variability (HRV) feature extraction pipeline using the PICS dataset.
- Construction of a multimodal neonatal pain classification framework integrating EEG spectral features, ERP features, ECG features, and respiration features extracted from the UCL dataset while integrating EEG, ECG, and respiration for the first time in a fully multimodal framework evaluated with strict LOSO cross-validation.
- Subject-independent evaluation using Leave-One-Subject-Out (LOSO) cross-validation to assess generalization across unseen neonates.
- Competitive classification performance using XGBoost, achieving AUC values of 0.929 and 0.849, demonstrating performance comparable with recent state-of-the-art methods.

II. RELATED WORK

A. Clinical Pain Assessment in Neonates

The assessment of pain in neonates remains a challenging clinical task because newborn infants cannot verbally communicate pain. Consequently, observational pain assessment scales have become the clinical standard for evaluating procedural pain in neonatal intensive care units (NICUs). Among these, the Premature Infant Pain Profile (PIPP) [1] and its revised version (PIPP-R) are the most widely adopted. These scoring systems combine behavioural indicators, including facial expressions and crying, with physiological responses such as heart rate, oxygen saturation, and gestational age to estimate pain intensity.

Despite their widespread clinical acceptance, observational scales possess several limitations. Pain scores depend heavily on trained observers and are therefore susceptible to inter-observer variability and subjective interpretation. Furthermore, manual scoring is labour-intensive and unsuitable for continuous bedside monitoring. Clinical assessment can also be influenced by gestational maturity, neurological development, and the administration of analgesics, motivating the search for objective physiological biomarkers capable of providing automated pain assessment.

B. Physiological Biomarkers for Neonatal Pain

Advances in physiological monitoring have enabled the investigation of objective biomarkers associated with neonatal

pain. Electroencephalography (EEG) provides direct measurements of cortical activity and has been shown to exhibit characteristic nociceptive event-related potentials (ERPs) following painful stimulation. The pioneering work of Slater *et al.* [7] demonstrated that noxious procedures evoke reproducible cortical responses even in premature infants, establishing EEG as an important modality for neonatal pain assessment.

Autonomic nervous system responses have also been extensively studied using electrocardiography (ECG) and respiratory signals. Pain-induced alterations in heart rate, heart-rate variability (HRV) [8], and respiration provide complementary physiological information regarding autonomic activation. However, each individual modality is susceptible to motion artefacts, developmental variability, and environmental influences. Consequently, recent studies have increasingly explored multimodal approaches that integrate cortical and autonomic responses to obtain a more comprehensive representation of neonatal pain.

C. Machine Learning for Automated Neonatal Pain Assessment

Machine learning has emerged as a promising approach for transforming multimodal physiological recordings into objective pain assessment tools. Early studies primarily focused on handcrafted EEG features, including event-related potentials and spectral characteristics, while subsequent work incorporated autonomic features derived from ECG and respiration.

The publicly available University College London (UCL) neonatal pain dataset has become one of the principal benchmarks for developing automated neonatal pain classifiers because it provides synchronized EEG, ECG, respiration, and clinically annotated PIPP scores collected during painful and non-painful procedures.

Among recent studies, Hamwi *et al.* [5] reported an AUC of 0.94 using Logistic Regression for older gestational-age infants. Similarly, Roué *et al.* [6] achieved an AUC of 0.90 using a Gradient Boosting classifier trained on the complete multimodal dataset. These studies demonstrate the effectiveness of combining physiological biomarkers with machine learning, while also establishing strong performance baselines for the UCL dataset.

D. Research Gap and Motivation

Although recent studies have demonstrated promising performance on neonatal pain classification, several important challenges remain. Existing approaches frequently focus on a limited subset of physiological features or employ evaluation strategies that may not adequately assess subject-level generalization. In addition, relatively few studies explicitly validate their signal preprocessing pipelines prior to large-scale model development.

The present work addresses these limitations through a two-stage methodology. First, the ECG preprocessing and feature extraction pipeline is validated using the PICS dataset to ensure reliable physiological signal processing. Subsequently, a multimodal feature extraction framework integrating EEG,

ECG, and respiration is developed and evaluated on the UCL dataset using Leave-One-Subject-Out (LOSO) cross-validation. This subject-independent evaluation protocol better reflects the practical clinical scenario in which the model is required to generalize to previously unseen infants.

III. MATERIALS AND METHODS

A. Datasets

1) *PICS Dataset (Pipeline Validation)*: The Pain in Infants Clinical Study (PICS) [9] dataset was employed exclusively for validating the proposed ECG preprocessing and feature extraction pipeline. The dataset consists of continuous ECG recordings collected from 10 infants during clinically required procedures. ECG signals were sampled at 500 Hz and processed using the NeuroKit2 [10] toolbox to perform signal cleaning, R-peak detection, and heart-rate variability (HRV)-analysis [8].

Unlike the UCL dataset, the PICS dataset was not used for classifier training. Instead, it served as an independent validation dataset to verify the robustness of the ECG preprocessing methodology prior to multimodal pain classification. Following preprocessing, a total of 26,208 ECG windows were successfully extracted and analyzed.

TABLE I
PICS DATASET OVERVIEW

Dataset	PICS
Subjects	10 infants
Signal	ECG
Sampling Rate	500 Hz
Windows Extracted	26,208
Purpose	ECG pipeline validation

2) *UCL Dataset (Pain Classification)*: The primary experiments in this study were conducted using the publicly available University College London (UCL) neonatal pain dataset [11] obtained through the UK Data Service. The dataset contains synchronized multimodal physiological recordings acquired during clinically relevant experimental procedures.

A total of 112 neonates were included in the study, resulting in 308 successfully processed recordings out of 310 available recordings. Each recording contains synchronized EEG, ECG, and respiration signals collected during heel lance (pain), sham control, and auditory control procedures.

Pain labels were generated using both experimental condition information and Premature Infant Pain Profile (PIPP) scores. In addition to the complete dataset, a PIPP-confirmed subset was constructed by selecting recordings with clinically significant pain scores ($\text{PIPP} \geq 7$), enabling a stricter evaluation protocol.

B. Overall Framework

The overall methodology adopted in this work is illustrated in Figure 1. The framework consists of five sequential stages: data acquisition, physiological signal preprocessing, multimodal feature extraction, feature matrix construction, and subject-independent pain classification using XGBoost [12] with Leave-One-Subject-Out (LOSO)-cross-validation [13].

TABLE II
UCL DATASET OVERVIEW

Dataset	UCL
Subjects	112 neonates
Processed Recordings	308 / 310
Signals	EEG, ECG, Respiration
Sampling Rate	2000 Hz (EEG)
Pain Stimulus	Heel Lance
Control Conditions	Sham, Auditory
Labels	Condition + PIPP Score
Purpose	Pain Classification

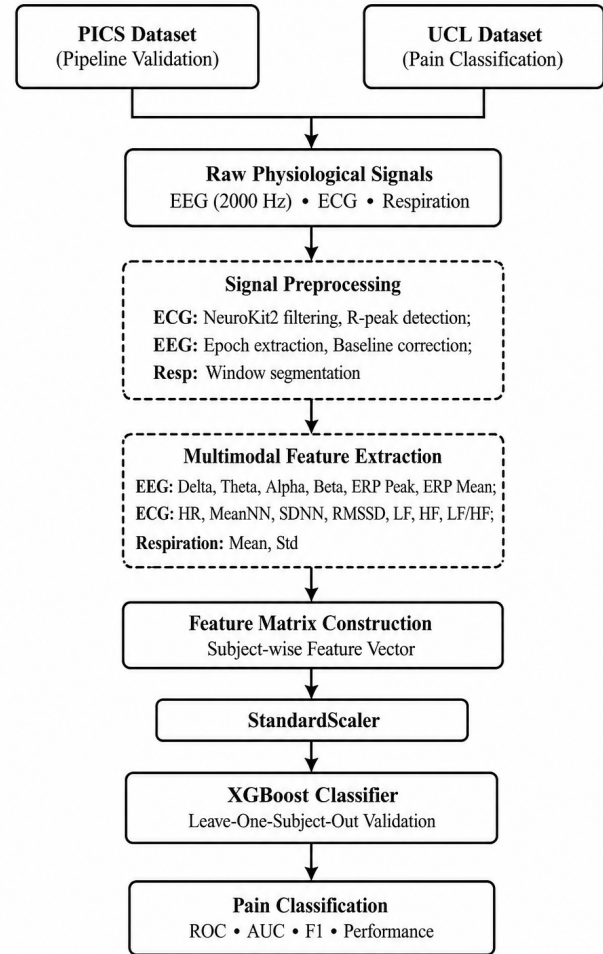


Fig. 1. Overview of the proposed framework.

C. Signal Processing

EEG preprocessing involved spectral analysis using Welch's power spectral density estimation [14], extraction of delta, theta, alpha and beta band power, computation of event-related potential (ERP) characteristics, and estimation of Global Field Power (GFP) [15]. The raw multimodal signals are illustrated in Figure 2

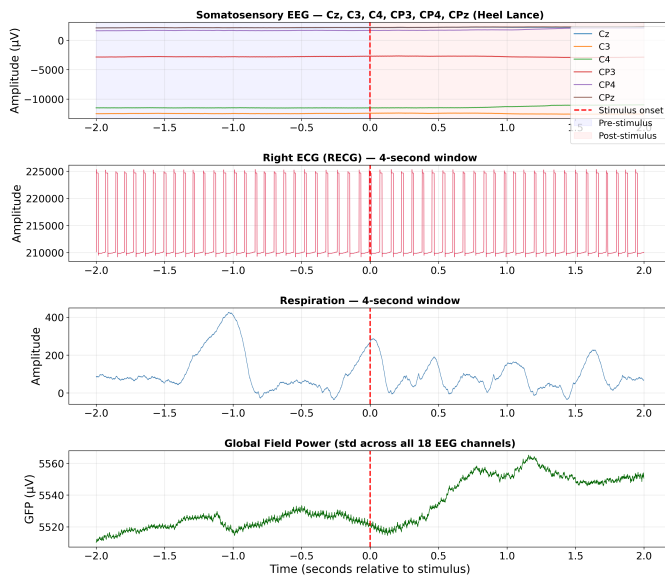


Fig. 2. Representative multimodal physiological recording from the UCL dataset showing synchronized EEG, ECG, and respiration signals aligned to stimulus onset.

D. Feature Extraction

Following modality-specific preprocessing, quantitative physiological features were extracted from the EEG, ECG, and respiration signals to construct a unified multimodal feature representation for pain classification. Following feature extraction, a total of 63 features were retained after removing columns with missing values exceeding 30%, yielding a unified feature vector per recording.

For the ECG modality, heart rate (HR) and Heart Rate Variability (HRV) features were computed from the detected R-peaks, including MeanNN, SDNN, RMSSD, pNN50, LF power, HF power, and the LF/HF ratio. EEG features comprised spectral band power (Delta, Theta, Alpha, and Beta) together with event-related potential (ERP) descriptors extracted from the post-stimulus response window. Respiration features were represented using statistical descriptors, including the mean and standard deviation within the analysis window.

Figure 3 presents an exploratory analysis of the ECG-derived physiological features extracted from the PICS dataset. The distributions of HR, RMSSD, and SDNN across individual infants demonstrate both inter-subject variability and consistent physiological trends, while the feature correlation matrix highlights the relationships among the extracted HRV metrics. This analysis provides additional validation of the extracted feature set prior to multimodal classification.

Table III summarizes the complete set of features extracted from each physiological modality and used to construct the final multimodal feature matrix.

E. Classification

Feature standardization was performed independently within each LOSO fold using StandardScaler to prevent information

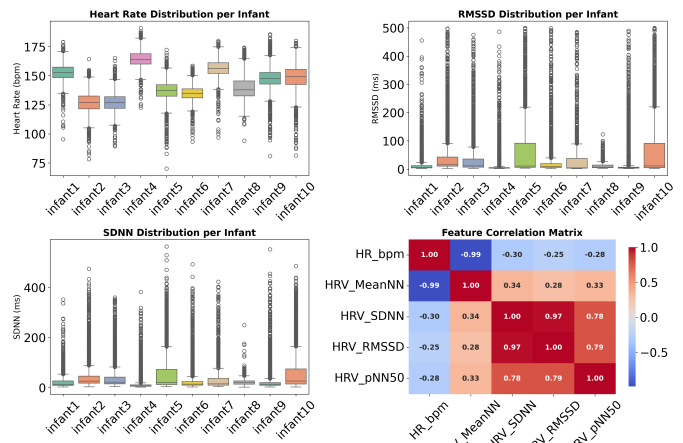


Fig. 3. Exploratory analysis of ECG-derived HRV features extracted from the PICS dataset. Box plots show inter-subject variability in heart rate, RMSSD, and SDNN across 10 infants, alongside the feature correlation matrix.

leakage. The XGBoost classifier was trained with class imbalance compensation via the `scale_pos_weight` parameter, set as the ratio of negative to positive training samples within each LOSO fold. The XGBoost classifier was configured with `n_estimators` = 300, `max_depth` = 3, `learning_rate` = 0.03, `subsample` = 0.8, and `colsample_bytree` = 0.8.

IV. EXPERIMENTAL RESULTS

Prior to multimodal pain classification, the proposed ECG preprocessing and feature extraction pipeline was independently validated using the PICS dataset. The objective of this experiment was to verify the robustness of the ECG processing methodology before its application to the larger multimodal UCL dataset.

Raw neonatal ECG recordings were processed using the NeuroKit2 framework for signal cleaning, R-peak detection, heart rate estimation, and heart-rate variability (HRV) analysis. Following preprocessing, a total of 26,208 ECG windows were successfully extracted from recordings collected across 10 infants.

Figure 4 illustrates a representative preprocessing example. The upper panel presents the cleaned ECG signal together with automatically detected R-peaks, while the lower panel shows the corresponding heart-rate profile derived from the detected cardiac cycles. The successful identification of R-peaks in neonatal ECG recordings, despite the presence of motion artefacts and physiological noise, demonstrates the robustness of the proposed preprocessing pipeline. The extracted HRV features subsequently formed the ECG component of the multimodal feature matrix used for pain classification.

A. Validation of the ECG Preprocessing Pipeline Using the PICS Dataset

The ECG preprocessing and HRV feature extraction pipeline was validated independently on the PICS dataset prior to multimodal classification. Raw neonatal ECG recordings from 10 infants were processed using NeuroKit2 for signal cleaning,

TABLE III
EXTRACTED FEATURES

Modality	Features
EEG	Delta, Theta, Alpha, Beta Band Power, ERP Peak, ERP Mean, Global Field Power
ECG	Pre/Post Heart Rate, Heart Rate Change, MeanNN, SDNN, RMSSD, LF, HF, LF/HF Ratio
Respiration	Mean, Standard Deviation

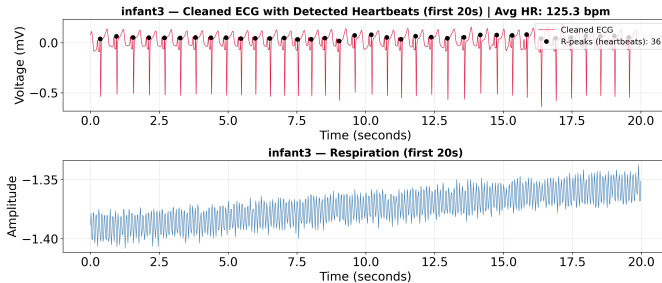


Fig. 4. Validation of the ECG preprocessing pipeline using the PICS dataset. The cleaned ECG waveform with automatically detected R-peaks demonstrates reliable heartbeat detection following signal preprocessing, enabling robust heart rate and heart rate variability (HRV) feature extraction.

TABLE IV
CLASSIFICATION PERFORMANCE USING LOSO CROSS-VALIDATION

Evaluation Group	Subjects	Mean AUC	Mean F1-score
Broad Condition-based	53	0.849 ± 0.285	0.475 ± 0.450
PIPP-confirmed	28	0.929 ± 0.175	0.452 ± 0.438

R-peak detection, and HRV analysis, yielding 26,208 ECG windows. Figure 4 illustrates a representative result: automatically detected R-peaks on cleaned ECG demonstrate reliable heartbeat identification despite motion artefacts, confirming pipeline robustness for neonatal signals.

B. Multimodal Pain Classification on the UCL Dataset

The proposed multimodal framework was evaluated on the UCL neonatal pain dataset using Leave-One-Subject-Out (LOSO) cross-validation to ensure subject-independent assessment. EEG, ECG, and respiration features extracted from each recording were combined into a unified feature vector and classified using an XGBoost classifier.

Two evaluation protocols were considered. The first used broad condition-based labels (heel lance versus control), resulting in 53 evaluable subjects. The second used a clinically validated subset containing recordings with available Premature Infant Pain Profile (PIPP) scores, where samples with $PIPP \geq 7$ were considered pain events, resulting in 28 subjects.

The quantitative performance under both evaluation protocols is summarized in Table IV.

Figure 5 compares the ROC curves obtained under both evaluation protocols. Both curves remain well above the random classification baseline, demonstrating reliable discrimination between pain and non-pain recordings. The PIPP-confirmed cohort achieved a higher mean LOSO AUC (0.929) than the broad condition-based cohort (0.849), indicating that

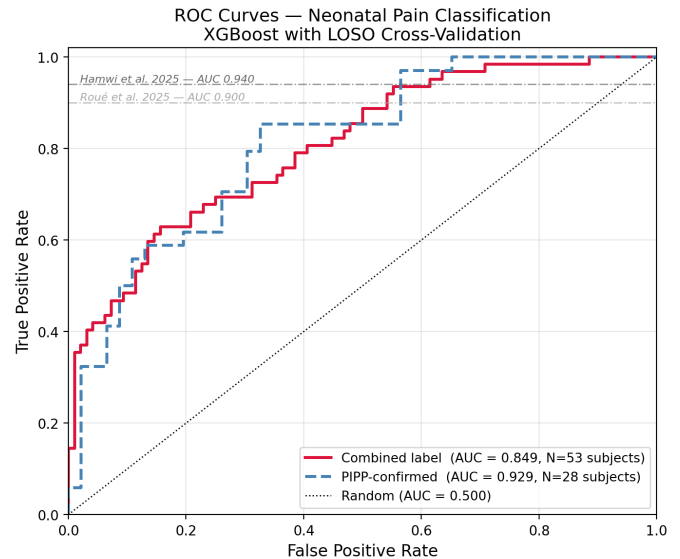


Fig. 5. Receiver Operating Characteristic (ROC) curves obtained using Leave-One-Subject-Out (LOSO) cross-validation for the proposed XGBoost classifier. The framework achieved a mean LOSO AUC of 0.849 using broad condition-based labels (53 subjects) and 0.929 using the clinically validated PIPP-confirmed subset (28 subjects). The diagonal dotted line denotes random classification, while horizontal reference lines indicate previously reported EEG-based performance.

TABLE V
COMPARISON WITH RECENT STUDIES

Study	Classifier	Validation	AUC
Hamwi et al. [5]	Logistic Regression	LOSO	0.94 [†]
Roué et al. [6]	Gradient Boosting	Subject-level	0.90
Proposed Method	XGBoost	LOSO	0.929

[†]Restricted to infants with gestational age ≥ 34 weeks.

clinically validated pain labels provide more discriminative supervision for multimodal learning.

Overall, the proposed multimodal framework demonstrates that combining cortical (EEG) and autonomic (ECG and respiration) biomarkers enables robust subject-independent neonatal pain classification, with performance comparable to recent state-of-the-art studies while using multimodal physiological information.

C. Comparison with Existing Methods

The proposed framework was compared with recent studies that employed the same UCL neonatal pain dataset. Table V summarizes the reported performance of representative machine learning approaches.

The proposed framework achieves performance comparable to recently published methods while employing a comprehensive multimodal feature set extracted from EEG, ECG, and respiration signals. Notably, although Hamwi *et al.* [5] reported a higher AUC of 0.94, that result was obtained exclusively on the oldest gestational-age cohort (≥ 34 weeks), where cortical pain responses are most mature and discriminable. The proposed framework achieves an AUC of 0.929 across all gestational-age groups under the same LOSO evaluation protocol, demonstrating that multimodal physiological integration enables robust subject-independent classification without restricting analysis to the most developmentally advanced subgroup. These results indicate that the proposed framework provides a reliable and clinically relevant approach for automated neonatal pain assessment.

V. DISCUSSION

The proposed framework demonstrates that multimodal physiological fusion improves neonatal pain classification over single-modality approaches. The two-stage pipeline — validated first on PICS and then applied to UCL — ensures that extracted features reflect genuine physiological signal content rather than preprocessing artefacts.

The performance gap between the broad condition-based (AUC 0.849) and PIPP-confirmed (AUC 0.929) cohorts reveals an important finding: ground-truth label quality directly determines classification ceiling. Procedure-based labels (lance versus sham) introduce noise because some heel lance recordings show low or absent pain responses (PIPP < 7), likely due to analgesic administration or neurological immaturity. Clinically validated PIPP labels filter these ambiguous cases, yielding cleaner class boundaries and higher discriminability.

The high standard deviations observed (AUC ± 0.285 and ± 0.175) reflect the inherent challenge of LOSO evaluation with small per-subject test sets of one to three recordings rather than true model instability. This variance is consistent with findings reported by Varoquaux [13] on cross-validation in small clinical samples.

Limitations: The evaluable subject count is constrained by PIPP score availability (79 of 112 subjects have scores) and the requirement for mixed-class test sets in LOSO. External validation on an independent neonatal cohort and interpretability analysis using explainable AI techniques remain as future work.

VI. CONCLUSION

This paper presented a multimodal machine learning framework for automated neonatal pain classification using EEG, ECG, and respiration signals. A two-stage experimental pipeline was employed: the ECG preprocessing and HRV feature extraction workflow was validated on the PICS dataset, and multimodal pain classification was evaluated on the UCL neonatal pain dataset comprising 308 recordings from 112 neonates using XGBoost with LOSO cross-validation.

The framework achieved AUC values of 0.849 and 0.929 under condition-based and PIPP-confirmed evaluation protocols,

respectively, demonstrating competitive performance relative to recent EEG-only approaches while incorporating complementary cardiovascular and respiratory information. These results confirm that clinically validated labels and multimodal physiological integration jointly improve subject-independent pain classification.

Future work will investigate model interpretability through explainable AI techniques, extend evaluation to additional neonatal cohorts, and explore real-time deployment for bedside clinical support.

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